Regression WiSARD application of controller on DC STATCOM converter under fault conditions

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***Abstract*—Capable of supplying local loads, DC microgrids have received much attention in the last decade for alleviating power flow through the main power grid. This has been achieved through the use of edge devices on the control of the converters, but, among other problems, microgrids have stability issues when Constant Power Loads (CPL) are present. This problem was already solved in the literature with the DC STATCOM power converter, in normal operation mode, it can deal with the grid operation. However, in fault cases, the solutions available still fail to ignore faults or even contribute to them. The present work aims to explore the potential of a light machine learning algorithm of the type Weightless Artificial Neural Network (WANN) for predicting the output of the original controller used in the DC STATCOM on an edge device connected to a converter, and investigate its generalization capability under microgrid fault situations. The WANN used is based on the regression variant of the Wilkes, Stonham, and Aleksander Recognition Device (WiSARD), coined as Regression WiSARD (ReW). The evaluation criteria employed measured the capability of the controller to reject the fault condition. Initial results showed surprisingly good results in comparison to the original DC STATCOM controller, indicating that a ReW-based controller plays well the role of the DC STATCOM and was able to cope with fault situations.**

***Index Terms*—weightless artificial neural networks, Regression WiSARD, smart grids, DC microgrids, edge computing.**

I. Introduction

Distributed Generation (DG) is a topic of increasing interest to the global energy sector. DC microgrid [1] constitutes a way to interconnect several DG's, as well as Energy Storage Systems (ESS), where dc loads can be also connected to. Each power converter heavily depend on edge devices since all data processing related to the converters must be performed locally. To further explore the fault handling capabilities of a machine learning-based converter, one must also comply with these requirements.

An example of dc microgrid is shown in Fig. 1, which comprise some DG, ESS, and local loads. Usually, a dc microgrids is connected to an ac power grid through a dc/ac converter know as Grid-Connected Converters (GCC) that can act as a grid former (if it provides voltage and frequency control), or as grid follower, depending on the power rating of that converter with respect to the installed power generation in the ac grid.

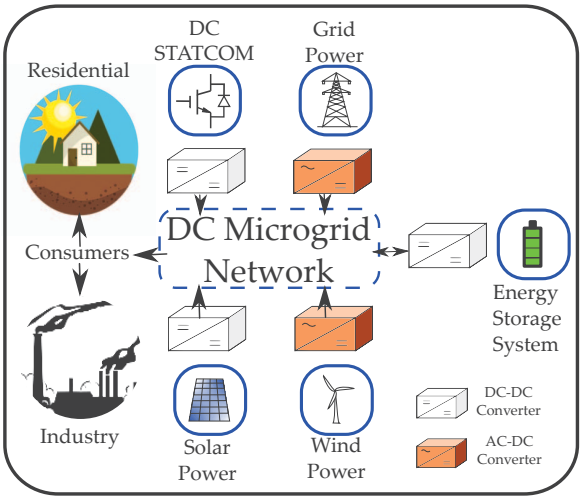


Fig. 1: DC microgrid connected to a conventional grid and to micro generators

A critical problem associated with dc microgrid control is the difficulty of controlling the dc bus voltage in the presence of dc loads that behave as Constant Power Loads (CPL) [2] [3], because the non linearity of these loads may render the system unstable. There are different ways of treating this problem, one of them is to employ a DC compensator associated with an energy storage system connected at the Point of Common Coupling (PCC). The use of such DC/DC compensator can bring benefits tho the DC microgrid, in analogy to the STATCOM and active filters that are capable of enhancing the quality of power, or improving the stability margin in AC grid/microgrids [4]. It is intuitive to assume that the same concept can be applied to DC microgrids. The Direct Current Static Compensator (DC STATCOM) was proposed by Chen [5] aiming at compensating theses instabilities on the DC microgrids, as depicted in Fig. 2.

However, DC STATCOMs can cause problems in case of fault in one of the loads, generators or transmission lines

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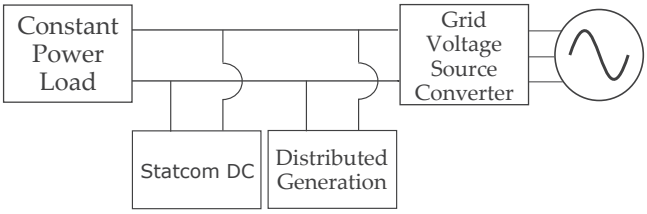


Fig. 2: DC STATCOM connected to a DC microgrid.

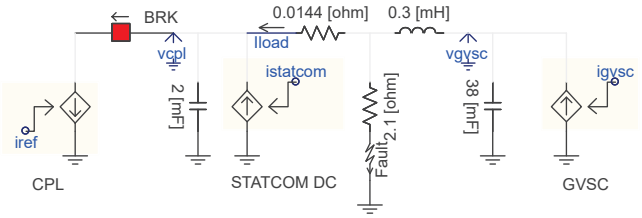


Fig. 3: DC microgrid modeled in PSCAD/EMTDC Environment

[6]. This happens when such converters try to compensate for unpredicted faults, causing damage to equipment or the collapse of the entire microgrid. The application of artificial neural networks (ANNs) based control to AC micro-grids has proved to improve the energy quality [7] during faults, but conventional ANNs are computationally heavy and difficult to adapt to concept drifts [8]. Therefore, this work proposes a DC STATCOM controlled by a lighter artificial neural network model, in particular a kind of weightless ANN (WANN) [9]. The investigation will be twofold; on one hand the capacity for energy quality improvement is evaluated and, on the other hand, the ability of the new mechanism to reject perturbations caused by faults. The remaining text is organized as follows: (i) Section II describes the DC microgrid modeling in PSCAD/EMTDC, the DC STATCOM modeling and their simulation; (ii) Section III explains the WiSARD variant employed (Regression WiSARD); (iii) Section IV explains the construction of the dataset; (iv) Section V presents experimental results with the ReW-based controller, and (v) Section VI concludes pointing to future research topics."

II. Modeling Adopted

A. DC microgrid modeling

In the development of a DC STATCOM, Chen [5] proposes a model for a microgrid [4], which in this work was implemented in the PSCAD/EMTDC environment. The modeled microgrid, as seen in the Fig. 3, consists of a GVSC, a CPL, and the DC STATCOM connected in the point of common connection(PCC). A square wave signal with an amplitude of 0.6 PU is given to the reference of the CPL, producing an intermittent power consumption.

Even though simple, the microgrid was modeled using virtual impedance techniques based on [2], accomplishing what was proposed in [2], a marginally stable grid. Figure 4 shows well the stability problem: a step of 0.6MW is applied

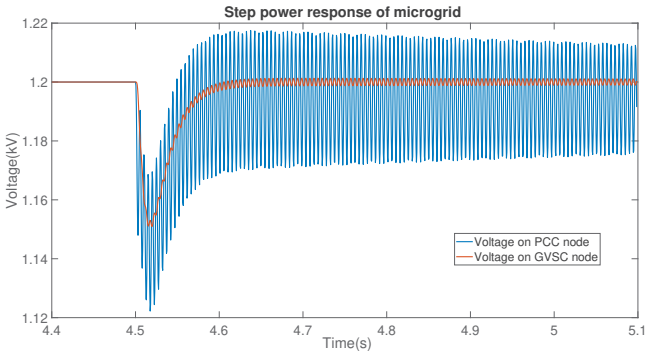


Fig. 4: Graphic of voltage on the PCC during activation of CPL

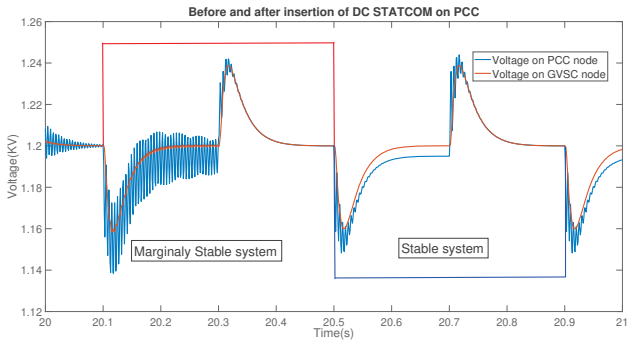


Fig. 5: STATCOM compensating the instabilities in the grid

on the CPL reference signal, causing the voltage on the PCC to oscillate, this way causing the DC grid to operate in a marginal stability state condition. It means that any rise in demanded power can cause the collapse of the grid or damage other loads connected in the grid.

B. DC STATCOM modeling

The instability problem of the microgrid comes from the non-linearities of the CPLs. To compensate these oscillations, there can be used as an energy compensator, like the DC STATCOM inserted in the PCC, as described in [5] and seen in Fig. 5.

Using the Eq. (1) taken from the [5] it was possible to implement the control for the DC STATCOM converter.

$$Reg(s) = \frac{1 + T_2 S}{(1 + T_1 S) R_v} \cdot \frac{T_f S}{1 + T_f S} \tag{1}$$

The implementation of Eq. (1) made in PSCAD/EMTDC can be seen in Fig. 6.

C. Study on transmission line faults

An important problem that affects microgrids are faults like short circuits in converters or the transmission line. As seen in Fig. 7 the microgrid becomes unstable even after the end of the fault. With the insertion of the DC STATCOM seen in Fig. 8, it can be seen a considerable improvement in the after

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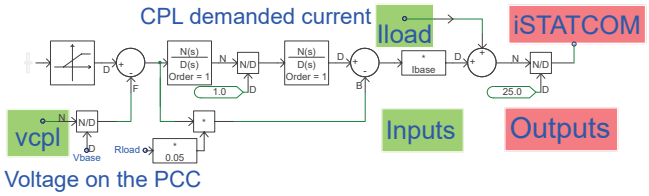


Fig. 6: Control diagram of DC STATCOM

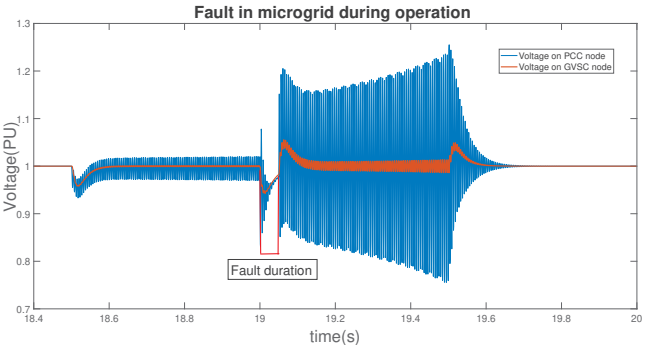


Fig. 7: Voltage on PCC and at GVSC without dc STATCOM connected during fault

fault energy quality, reducing the voltage overshoot more than 10% in global measurements.

Although the impressive improvement in the energy quality after fault, there is still exist room for improvement. Since even though the DC Statcom compensates most of the oscillation in the PCC, its control is still governed by a linear controller, and with this suffers from the instability of the fault on the grid.

It's already well consolidated that ANN controlled converters improve significantly transient stability of AC power grids [10]. That way, it's expected that the same applies to the direct current case, but at a cost of expensive processing power. One possible solution for the problem is the use of WANN controllers because they would be capable of mapping the fault condition, compensate them better, and since WANN

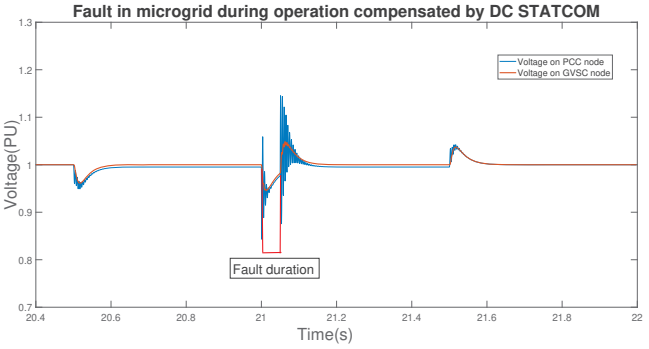


Fig. 8: Voltage on PCC and at GVSC with dc STATCOM connected during fault

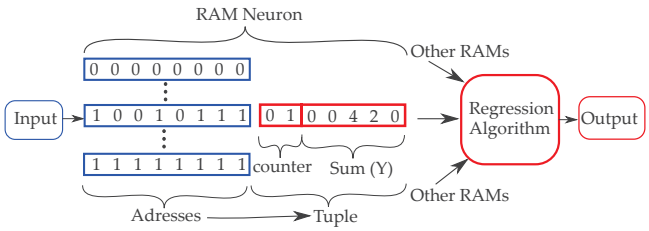


Fig. 9: RAM Neuron model of the ReW

doesn't have multiple layers and as it doesn't need convergence techniques, it is way easier to implement it computationally [11].

III. Regression WiSARD

Unlike mainstream models of artificial neural networks, which are based on the biological functioning of synapses connected to the neuron soma, Weightless Artificial Neural Networks (WANN) are based on dendritic trees [12], the most notable morphological structure of the neuron. For this, WANNs are based on Random Access Memory(RAM)-nodes, so that the process of learning a WANN consists only of memory writing, while its classification phase consists only of memory reading.

One of WANN's simplest and least costly models is called WiSARD [13], a n-tuple classifier characterized by its fast training time. For dealing with regression tasks, WiSARD received an extension called Regression WiSARD (ReW), that was idealized in [14] as a lightweight substitute for the more computationally costly Artificial Neural Networks available.

A. ReW-neuron model

The traditional neuron of WiSARD consists of a piece of RAM, the contents of each memory location being an access counter. The neuron of the ReW extension is a RAM node where each memory location contains a counter and a "partial" $\sum y$ result, formed by the sum of the prediction values of the previous learned samples. Both values are updated at each new training pattern, and all begin with zero initial values, as seen in Fig. 9.

B. ReW - Input

The retina is equivalent to the input layer of a neural network, and is formed by the processed data that will be used for training and classification. It has a fixed size of $N * n$ where N is the size of the address and n is the size of the RAM Node, as exemplified in Fig. 10.

The processed data must be in the binary form to be in the retina, but it may pass for various preprocessing steps before that.

C. ReW-neuron mapping

n bits in the retina are used to map the addresses of a RAM node in a pseudo-random way, where various positions of the data in the retina are randomly selected in the first training sample and used to map the address of the RAM Node.

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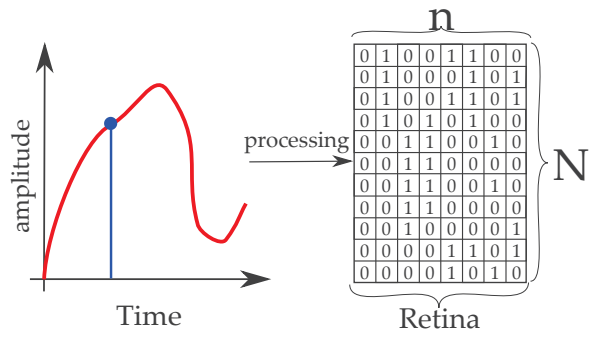


Fig. 10: Retina model

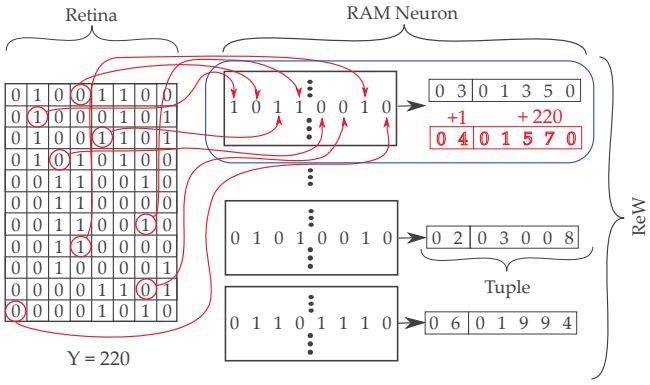


Fig. 11: Training of the ReW

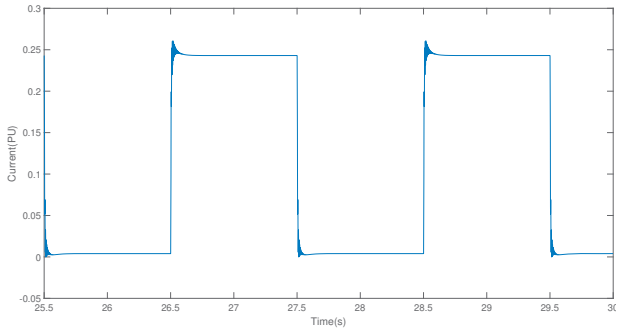


Fig. 12: Current waveform in the DC link under normal operation.

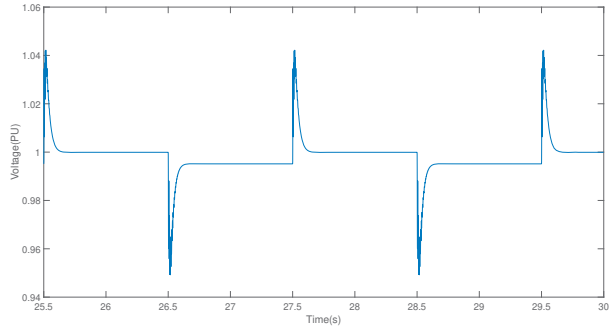


Fig. 13: Voltage waveform in the DC link under normal operation.

D. Training phase

During this step, the network is presented with k pairs(X_i, Y_i) which update the counter and the Sum Y with the new Y_i data as demonstrated in Fig. 11.

Lusquino shows in [14] that ReW presents a minimal difference to the state of the art ANNs, surpassing its speed in many orders of magnitude, in training and prediction phases.

E. Prediction phase

During this step, the sum of counters($\sum c$) and partial y ($\sum y$) of the position accessed by a given **x** are used to calculate the corresponding y, given a regression function like the following:

Simple mean: $\frac{\sum y}{\sum c}$

Power mean: $(\frac{1}{n}\sum_{k=1}^{n}(y_k^{c_k}))^{\sum c}$

Median: central value of $\frac{y_i}{c_1}$

Harmonic mean: $(\frac{\sum_{k=1}^{n} y_i^{-1}}{n})^{-1}$

Harmonic power mean: $(\frac{n}{\sum_{i=0}^{n}\frac{c_i}{y_i}})$

Geometric mean: $(\prod_{i=0}^{n}\frac{y_i}{c_1})^{\frac{1}{n}}$

Exponential mean: $\log(\frac{\sum_{i=0}^{n} e^{\frac{y_1}{c_i}}}{n})$

It is also possible to ignore the contribution of an address that does not contain a minimum of "0"s or "1"s.

IV. Dataset Creation

The preprocessing of data consists of most of the logistic and computational effort when talking about neural networks, so that is necessary to analyze the available data and present them the best way to the neural network.

All data used in the present paper can be found in the following link link.

A. Data extraction

The data was extracted from the software PSCAD, where it is possible to establish which signal and what interval these data will be exported, given the possible options the easiest was to export in CSV and opening in the PYTHON environment.

The extracted data was: the voltage on the PCC, the current come from GVSC and the current provided by the DC STATCOM under normal operation mode. One small segment of data can be seen in the Figs. 12, 13 and, 14 correspondently. Data was divided into three parts, two-thirds for the training and one-third for validation. It was also extracted the same data on fault condition, where it will be used to observe how the WANN behave.

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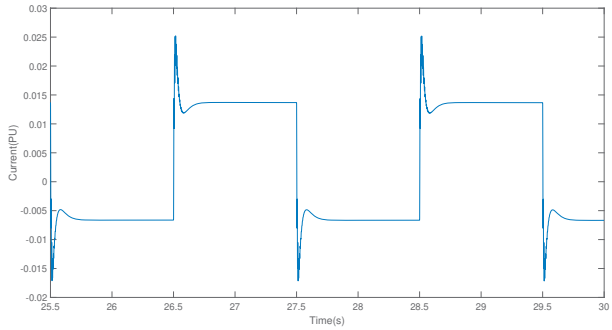


Fig. 14: Current waveform in the output of DC STATCOM under normal operation.

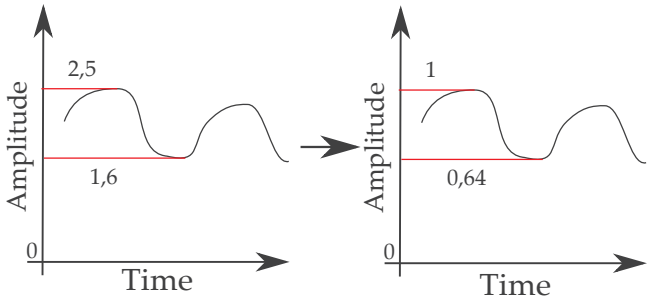


Fig. 15: Normalization demonstration

B. Data Processing

To use the extracted data on the WANN it was necessary to preprocess it, as to proportionate data in the form the ReW was projected to receive as input.

This way the data treatment was done in the following manner:

- Normalization
- Offset
- Discretization
- Binarization
- Gray code conversion
- Signal windowing

a) Normalization: Observing the extracted data it was possible to measure the maximum variation of parameters, with this the data were constrained between 0 and 1, as demonstrated in Fig. 15. The voltage and current were normalized in different forms so they occupied the maximum possible of the spectrum of possible values during discretization.

b) Offset: By having negative values on the current signal during transitions, it was necessary to offset the data so all the values exist on the natural number domain as demonstrated in Fig. 16.

c) Discretization: With the normalized data ready, the discretization was done to fit the data on an interval from 0 to 2^B, where B is the resolution of the discretization chosen.

d) Binarization: After the data were distributed into the interval of choice, it was only needed to binarize the data to bring it to the digital world as demonstrated in Fig. 17.

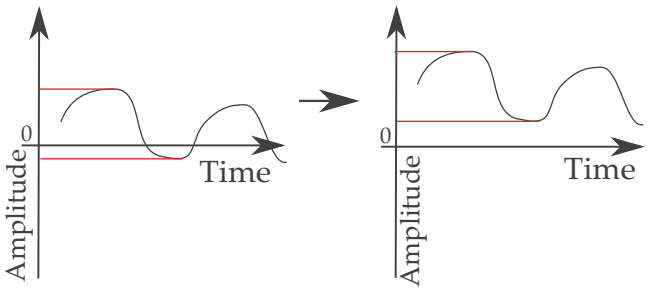


Fig. 16: Offset demonstration

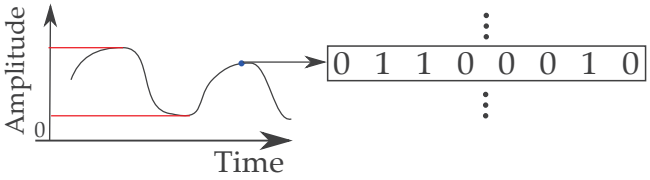


Fig. 17: Binarization demonstration

It was necessary to maintain the length of the binary numbers so they could be later arranged in the dataset.

e) Gray code conversion: It was decided that to better take advantage of the ReW, it was necessary to convert the binarized data into Gray code [15]. This was done because the data regression on a WANN, assumes that the entry values must be similar since they will be used as pseudo-aleatory addresses. This way the Gray code does a better job to represent close numbers.

f) Signal windowing: Signal windowing consists of the selection of N time-delayed samples of the voltage and current signals already converted in Gray code and laying the sample data in a matrix form where the x-axis is the gray code data of one sample and the y-axis is the N time-delayed samples with the most recent first and the oldest samples coming along. This way each signal can be converted to T matrices of dimension $B \times \mathrm{N} where \mathrm{T} is the total of points in the signal.$

g) Data-base / Retina: For the creation of the dataset used in this work, it was established that the length of the retina would be the binary size of the discretization of the data signal and the height would be two times the number of delayed samples where the first half was the voltage signal and

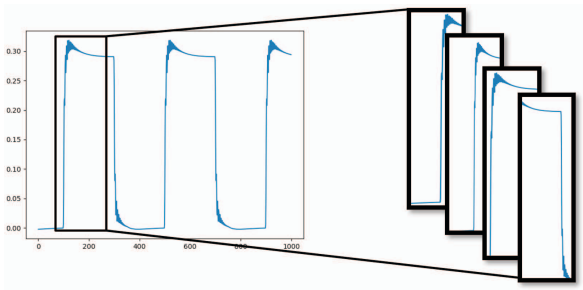


Fig. 18: Signal Windowing of the Current signal

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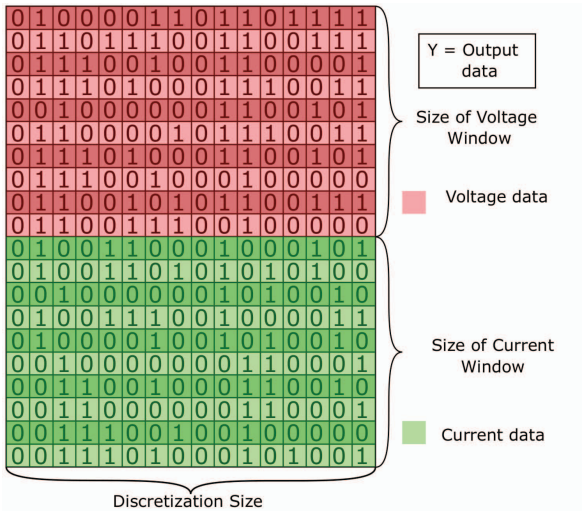


Fig. 19: Retina structured for the training of the ReW

the second half would be the current signal as exemplified in the Fig. 19.

In the end, each sample is a numerical value that refers to the current signal of the DC STATCOM obtained in the simulation, after doing this procedure to every point in the data we had an approximate of 10000 examples to train the ReW, a quantity that would prove to be enough to produce satisfactory results.

V. Experimental Results

A. Hyperparameters Search

After processing the data as mentioned in the last section IV, the ReW-based controller model was trained. However, it was observed that changing some parameters, like resolution (B), window size (N), and the address size (n) of the ReW controller made the performance of the model, and quality of the results, changed considerately. By doing a naive search within the hyperparameters, poor results were obtained, with MSE around 10%, being the best results observed, when the hyperparameters was set in the interval of 5 and 30.

As the space search for these hyperparameters was of the order of 25^3 which means 15625 possibilities, a Simulated Annealing (SA) hyperparameter search algorithm [16] [17] was used to identify a good estimation of the optimal hyperparameters that should be used.

Initially developed in 1983, the Simulated Annealing [17] algorithm is a metaheuristic search algorithm, that uses stochastic perturbations proportional to a descending temperature to change the seed parameters. The initial temperature used was 10 and it was implemented an exponential decaying temperature function to change the perturbation on the 3 hyperparameters chosen, witch was: resolution, window size, and address size. These seed parameters carry the configurations to set up the ReW for training, thus leading to a configuration where the ReW is well trained enough to predict the output with an acceptable error margin and operates with a reasonable

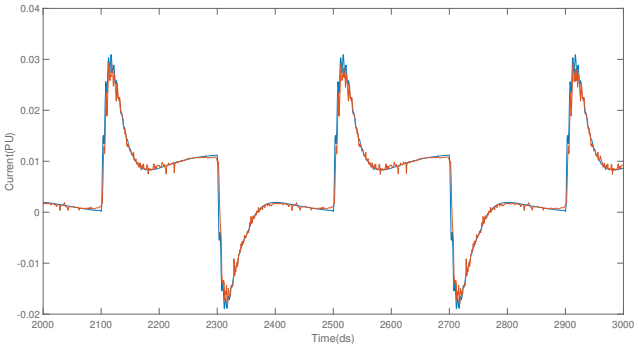


Fig. 20: Predicted data by ReW.

amount of computational power. The cost function chosen was the Mean Square Error(MSE), and by comparing the result of the ReW prediction with the training data, it's possible to measure the difference between the prediction and the already existing literature controller.

This difference between existing data and predicted data is used to calculate if the ReW improves or not with the perturbations on its hyperparameters [16]. By applying the SA algorithm it was possible to determine a good estimation for the optimum hyperparameters as the resolution being 14 bits, then window size as 6 time step delays, and the address size of the ReW as 13 bits. These results were used in the training of ReW, and provided a better understanding of how the parameters affects the end result of the prediction.

B. Training

Using the hyperparameters obtained with the SA algorithm, it was possible to train the ReW with the processed training data. The training batch would consist of $\frac{2}{3}$ of the total of samples obtained and the validation consisted of the remaining third. In total there where 9993 training samples and 4997 test samples used in the training and validation of the ReW. As mentioned in IV-B0g the retina is composed of current and voltage signal discretized were used as input and the corresponding current of DC STATCOM was used for the Y output.

As a note, it was observed that fewer samples could be provided by maintaining nearly the same quality of results.

C. Validation

After the training of the ReW, it was observed that the WANN was capable to generalize in a very satisfactory way the ideal current of the original converter. This can be seen in Fig. 20.

There can be observed in table I, for the ten rounds, trained and predicted with the same data but shuffled, the ReW obtained a mean MSE of 1.206 what shows that the network is capable to model the behavior of the controller. Another relevant info is that the variance of the error maintained small, proving the solidity of the ReW. The result of the predicted data can be used as a current reference to the DC STATCOM.

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TABLE I: MSE Table for individual run

Run	MSE(%)
1	1.32
2	1.19
3	1.12
4	1.23
5	1.16
6	1.15
7	1.11
8	1.30
9	1.24
10	1.21
10 runs mean(%)	1.21
Variance	4.464x10^{-07}

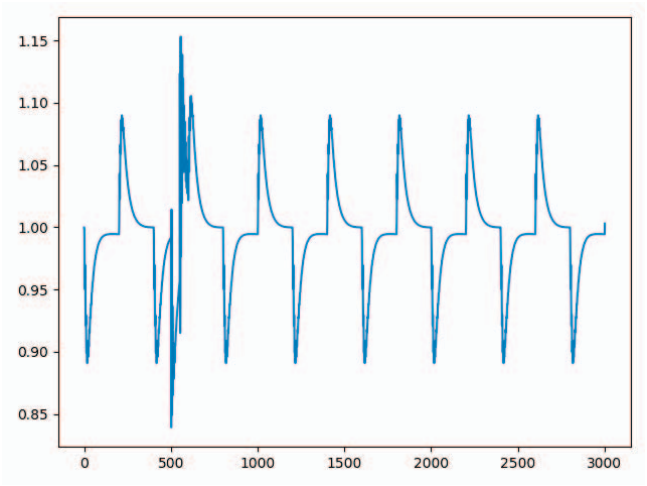


Fig. 21: Fault current in PCC

D. Test

A portion of the dataset that it would not have seen before was presented to the trained ReW. This portion of the dataset contained currents and voltages of a microgrid operating under fault condition. As seen in Figs. 21 and 22, the microgrid suffers a oscillation of 15% in voltage and of 100% in current. This condition cannot be fully compensated, but the ReW can help by bringing the microgrid to its normal operational condition faster than the conventional method.

During the fault condition the WANN controller minimize the contribution of the current going to the fault, which as seen in Fig. 23, improve the stability of the microgrid and the converter life spam.

Due to the implementation of the WANN, it could be observed that the ReW tries to approximate the output with the data it already has, to produce an output, that even tough noisy, it doesn't contribute to the short circuit current. Another less relevant remark is that the output signal recovers faster than the original signal.

VI. CONCLUSION

This work proposed a microgrid with an integrated DC STATCOM controlled by a lightweight machine learning algorithm that compensates for the instabilities caused by CPLs. In order to do so, it was implemented a microgrid with integrated DC STATCOM in the PSCAD/EMTDC environment. The preprocessing of the data for Regression WiSARD's training was performed and a study on the behavior of the trained ReW under faults condition was realized. As demonstrated, the WANN is viable to substitute the control in a DC STATCOM Converter. Also, given the processing speed of the ReW, there is a significant possibility of implementing the ReW on a low power processing unity as Digital Signal Processors (DSPs). Even though still requiring a deeper study of this kind of controller behavior in fault cases in DC microgrids, there are promising chances of obtaining good results. Further exploration of our preliminary results lies the following topics: (i) studying the capacity of generalization of the WANN in cases where the memories were not trained; (ii) converting the Regression WiSARD for shipping in DSPs; (iii) implementing

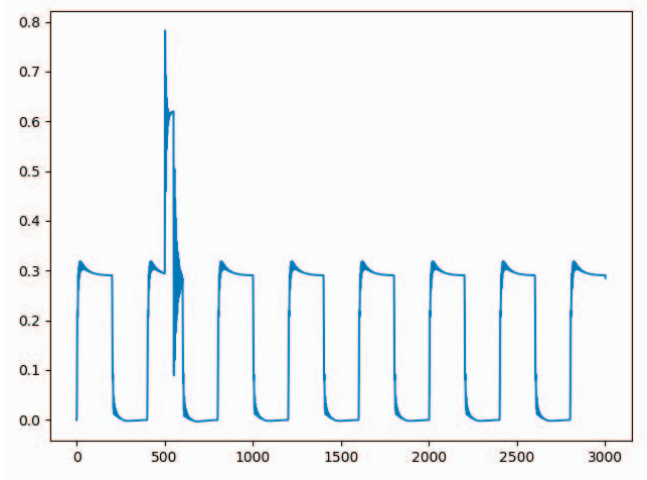


Fig. 22: Fault voltage in PCC

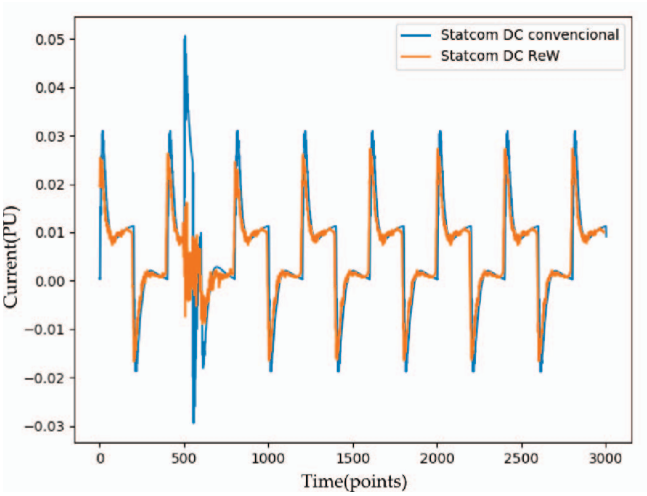


Fig. 23: Current data of the DC STATCOM with ReW and conventional case

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real-world DC STATCOM with ReW and connecting it to a real microgrid, in an integrated fashion, and (iv) evaluating fault (short circuit) situations and training WANN in these cases.

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